Seminar 1

Preferences, utility and counterexamples

This worksheet accompanies Lecture 1. No demand calculus is needed. Give definitions before calculations, and justify every answer. For a false universal claim, one fully specified counterexample is enough. The final exercise is reserve material.

Conventions. Rational means complete and transitive. Continuity means a closed weak-preference graph: if $x^n \to x$, $y^n \to y$ and $x^n \succeq y^n$ for all n, then $x \succeq y$. Convex preferences have convex upper contour sets. Strict convexity uses distinct bundles and mixtures with weights in $(0,1)$. Monotonicity means $x \gg y \Rightarrow x \succ y$; strong monotonicity means $x \geq y$, $x \neq y \Rightarrow x \succ y$. Local nonsatiation (LNS) means every relative neighbourhood of every bundle contains a strictly preferred bundle. All topological statements use the stated domain's relative Euclidean topology.

1. Relations before utility

Core · 15 minutes

On $X = \mathbb{R}^2_+$, classify each relation as complete or incomplete, transitive or intransitive, continuous or discontinuous. Prove every positive claim; give a numerical witness for every failure.

(a) $x \succeq_A y$ iff $x_1 \geq y_1$ and $x_2 \geq y_2$.

(b) $x \succeq_B y$ iff $x_1 \geq y_1$ or $x_2 \geq y_2$.

(c) $x \succeq_C y$ iff $x_1 + x_2 > y_1 + y_2$, or both sums are equal and $x_1 \geq y_1$.

Is relation C just an ordinary linear utility with a sufficiently small positive weight added to good 1?

2. True or false?

Core · 15 minutes

(a) If two bundles are incomparable, the consumer is indifferent between them.

(b) If a relation is represented by a real-valued function, it must be rational.

(c) If a representing utility is discontinuous, the preferences must be discontinuous.

(d) A convex utility function represents convex preferences.

(e) Strictly increasing transformations preserve convexity of preferences, but need not preserve concavity of utility.

(f) LNS implies that utility is unbounded above.

3. A kink, a transformation and a curve

Core · 20 minutes

Let $X = \mathbb{R}^2_+$, $u(x) = \min\{2x_1, x_2\}$ and $v(x) = e^{u(x)}$.

(a) Draw the indifference set through $(1,2)$ and its weak upper contour set. Include the corner and both rays.

(b) Are the preferences rational, continuous, convex, strictly convex, monotone, strongly monotone, locally nonsatiated?

(c) Prove that u is concave. Is v concave? Does your answer change the shape of the preferences?

(d) Is $w(x) = \mathbf{1}\{u(x) \geq 2\}$ another representation of the same preferences?

4. What exactly did we observe?

Core · 15 minutes

A consumer has a fixed weak ranking over $\{a, b, c\}$. In one observation they select a from $\{a, b\}$; in another they select b from $\{a, b, c\}$.

(a) If each observation records only one selected maximiser, do these observations refute rational preferences? Construct a ranking if they do not.

(b) Now the records are the *full* best-alternative sets: $C(\{a, b\}) = \{a\}$ and $C(\{a, b, c\}) = \{b\}$. Can a rational ranking generate them? State the contradiction.

(c) Which additional observation or recording convention would distinguish these interpretations?

5. Why does the budget line matter?

Core · 15 minutes

Let $B = \{x \in \mathbb{R}^2_+ : p \cdot x \leq m\}$, $p \gg 0$, $m > 0$.

(a) Assuming a best bundle x^* exists and preferences are LNS, prove $p \cdot x^* = m$ without derivatives.

(b) Does budget exhaustion at every best bundle imply LNS? Try $u(x) = -[(x_1 - 2)^2 + (x_2 - 2)^2]$ and $p = (1, 1)$, $m = 1$. Here "every" refers to every optimum for this one budget.

(c) Does LNS alone guarantee that a best bundle exists? Try a lexicographic ranking that first maximises x_1 and, conditional on a tie, minimises x_2, on B with $p = (1, 1)$, $m = 1$. Does *this* example actually fail existence? If not, repair the example using $X = [0, 1]$.

6. Can a finite grid repair lexicographic utility?

Reserve · 10 minutes

On $X_N = \{0, 1/N, \ldots, 1\}^2$, lexicographic preferences first compare x_1, then x_2. Find M for which $u_N(x) = Mx_1 + x_2$ represents them. Can the same finite M work on all of $[0, 1]^2$? Can *any* real-valued utility represent lexicographic preferences on the square?